Neural Caption Image Retrieval

Two page project write up | 30 September 2026

Problem statement

Given a natural language caption, retrieve and rank the matching image from a gallery. The task links sentence meaning to visual content, although captions and images begin in different feature spaces. The project compares a linear caption to image mapping with recurrent models that learn a shared space for both modalities.

Dataset

The experiments use MS COCO 2014 images and the Karpathy split: 113,287 training images, 5,000 validation images, and 5,000 test images. Images usually have five human captions; a few source records contain extras. Training captions define the vocabulary; validation and test captions do not influence it. Each image is resized to 256 by 256 pixels, represented by ten 224 by 224 crops including horizontal flips, and passed through VGG 19 pretrained on ImageNet. The ten 4,096 dimensional fc7 vectors are averaged. For pretrained text features, the project uses 300 dimensional GloVe 6B word vectors.

Approach

Linear baseline. Average available GloVe vectors in each caption, then fit a multiple response least squares map from 300 dimensional caption vectors to VGG image features. Rank images by squared Euclidean distance from the mapped query.

Neural models. Project VGG features into a 1,024 dimensional space. Encode caption words with either a GRU trained from scratch, a GRU initialized with GloVe, or an LSTM initialized with GloVe. The recurrent final state is the caption representation. Similarity is the negative squared Euclidean distance between the image and caption representations. A bidirectional margin ranking loss (margin 0.1) separates matching image caption pairs from mismatched pairs; other captions of the same image are not treated as negatives.

Training. The specified starting configuration uses Adam, batch size 128, up to 50 epochs, and early stopping based on validation. The paper lists a learning rate of 0.05; after an ineffective first full GRU epoch at that rate, training was restarted at 0.0001. The models use the same train/validation/test split and image features to keep the comparison aligned.

Retrieval path

For a new sentence, the system applies the training tokenizer and encodes its words with the selected caption model. It compares that representation with projected VGG features from the saved test gallery, sorts by negative squared distance, and returns the top ranked image paths and scores. The expensive VGG feature extraction is performed before retrieval, rather than once for every query.

Implementation overview

The repository prepares COCO manifests and vocabulary, extracts frozen VGG 19 features, fits the baseline, trains each recurrent variant, evaluates retrieval, and displays ranked images. Feature arrays are mapped into memory, and extraction and training can resume after interruption. The README gives setup and run commands.

Evaluation

Caption to image retrieval is evaluated on a fixed gallery of 1,000 test images, with one selected caption per image for each standard method (1,000 queries). Reported measures are Recall@1, Recall@10, and mean image rank. The pooled caption baseline combines the first five held out captions for each image into one query and is identified separately from single caption retrieval. Additional analysis checks recall by caption length, recall across training epochs, and short/long query examples.

Validation and example

Best validation Recall@10 was 75.2% (GRU), 78.3% (GRU + GloVe), and 76.8% (LSTM + GloVe); training stopped after 18, 16, and 14 epochs. For the test query 'a guy riding a bike next to a train,' GRU + GloVe ranked the matching COCO image first. The experiment figures show Recall@10 by epoch, caption length groups, and short and long examples.

Results

Method	R@1	R@10	Mean rank
Average GloVe baseline	4.5%	21.8%	170.5
Baseline + pooled captions*	4.9%	24.2%	158.8
GRU	24.8%	72.4%	16.6
GRU + GloVe	23.8%	76.1%	12.3
LSTM + GloVe	21.0%	72.0%	14.8

* Pooled caption baseline combines five test captions per image; other methods use one selected caption per image.

Conclusions

On the fixed 1,000 image gallery, the GRU with GloVe achieved the highest single caption Recall@10 (76.1%). The GRU without pretrained vectors achieved the highest Recall@1 (24.8%). The pooled caption baseline has extra query information, so its comparison is qualified. Conclusions apply to this split and evaluation protocol.

Code: github.com/kishansaaai/ML_PROJECT. Source: Qian and Lamberti, *Neural Caption Image Retrieval* (CS229).